

Advanced Optimization

Chapter 10 — Least-Squares Problems

Hongseok Kim

Department of Electronic Engineering, Sogang University

Fall 2026

Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 10

Why least squares gets its own chapter

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} \sum_{j=1}^m r_j^2(x) = \frac{1}{2} \|r(x)\|^2, \quad r: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad m \geq n. \quad (10.1)$$

With the Jacobian $J(x) = [\partial r_j / \partial x_i]$ ($m \times n$, rows ∇r_j^T):

$$\nabla f(x) = J(x)^T r(x), \quad \nabla^2 f(x) = \underbrace{J(x)^T J(x)}_{\text{first order in } r} + \underbrace{\sum_{j=1}^m r_j(x) \nabla^2 r_j(x)}_{\text{needs residual Hessians}}. \quad (10.4-10.5)$$

The whole chapter in one observation

Once J is computed — which the gradient already requires — the leading Hessian block $J^T J$ is *free*. And the second block is weighted by the residuals r_j , so it is *small* precisely when the fit is good or the model is nearly affine. Every algorithm here exploits one or both facts.

Where you meet (10.1): data fitting and parameter estimation everywhere; power-system state estimation (weighted nonlinear LS); and the training loss of a network with squared error is exactly this f .

Background (§10.1): the model, and why the 2-norm

Example 10.1. Blood-concentration model $\phi(x; t) = x_1 + tx_2 + t^2x_3 + x_4e^{-x_5t}$, measurements (t_j, y_j) , residuals $r_j(x) = \phi(x; t_j) - y_j$ (10.8); minimize $\frac{1}{2} \sum r_j^2$ — the sum of squared *vertical* deviations (Figure 10.1). “Fixed-regressor”: the t_j are exact, the y_j are noisy.

Other norms exist

$\max_j |r_j| = \|r\|_\infty$, $\sum_j |r_j| = \|r\|_1$ (10.11) — nonsmooth, but reformulable as smooth *constrained* problems (Ch. 17). ℓ_1 is the robust choice under outliers.

Why ℓ_2 is canonical

If the discrepancies ϵ_j are i.i.d. Gaussian, the likelihood (10.12) is $\propto \exp(-\frac{1}{2\sigma^2} \sum r_j^2)$: **least squares = maximum likelihood under Gaussian noise**. Different noise model $\Rightarrow$ different objective (weighted LS for known variances: $r^T W r$).

Statistics decides the objective; this chapter only minimizes it.

Linear least squares: three factorizations, one warning

$r(x) = Jx - y$ with constant J : $f(x) = \frac{1}{2}\|Jx - y\|^2$ is *convex*; $\nabla f = 0$ is sufficient (Thm 2.5) and reads

$$J^T J x^* = J^T y \quad (\text{the \textbf{normal equations}, 10.14}).$$

Assume $m \geq n$, full column rank.

Method	Work	Robustness	Use when
Cholesky of $J^T J$ (10.15)	cheapest	$\kappa(J^T J) = \kappa(J)^2$	$m \gg n$; $J^T J$ storable, J not
QR of J (10.17–10.18)	moderate	works with $\kappa(J)$	the default
SVD $J = U_1 S V^T$ (10.19)	most expensive	best; gives sensitivity	near rank-deficiency

QR: $\|Jx - y\|^2 = \|R(\Pi^T x) - Q_1^T y\|^2 + \|Q_2^T y\|^2$ — a triangular solve kills the first term; the second is the residual, untouchable by x .

The SVD bonus (10.21–10.22)

$x^* = \sum_{i=1}^n \frac{u_i^T y}{\sigma_i} v_i$: small σ_i flags a hypersensitive direction. Rank-deficient: zero those terms for the minimum-norm solution; dropping small- σ_i terms trades bias for stability (truncated SVD — a first meeting with regularization). Very large: CG on the normal equations, done right = LSQR (products with J , J^T only).

Gauss–Newton: Newton with the free half of the Hessian

Line-search Newton, but solve

$$J_k^T J_k p_k^{\text{GN}} = -J_k^T r_k \quad (10.23)$$

instead of $\nabla^2 f_k p = -\nabla f_k$ — i.e. use $\nabla^2 f_k \approx J_k^T J_k$ (10.24).

Four advantages

- 1 **No residual Hessians.** J_k came with the gradient; the approximation costs *zero* extra derivative evaluations.
- 2 **Often nearly Newton.** When $|r_j| \|\nabla^2 r_j\|$ is small next to the eigenvalues of $J^T J$ — small residuals or nearly affine ones — the neglected term is negligible and the rate is Newton-like.
- 3 **Always a descent direction** (full-rank J_k , $\nabla f_k \neq 0$): $(p_k^{\text{GN}})^T \nabla f_k = -\|J_k p_k^{\text{GN}}\|^2 < 0$ (10.25).
- 4 **The step is itself a linear LS problem:**

$$p_k^{\text{GN}} = \arg \min_p \frac{1}{2} \|J_k p + r_k\|^2 \quad (10.26)$$

— solve by QR/SVD/CG **without ever forming $J_k^T J_k$** (and without squaring κ).

Gauss–Newton, operationally

- ▶ **The other reading of (10.26).** Linearize the residual, $r(x_k + p) \approx r_k + J_k p$, and minimize the norm of the linear model: GN is “model the *residual* linearly”, where Newton is “model the *objective* quadratically”. Same step, better numerics.
- ▶ **Step length:** Wolfe or Armijo from Ch. 3, trying $\alpha = 1$ first.
- ▶ $m \gg n$: accumulate $J^T J = \sum_j (\nabla r_j)(\nabla r_j)^T$ and $J^T r = \sum_j r_j \nabla r_j$ residual by residual (10.27) — J itself is never stored.
- ▶ **Large and sparse:** inexact GN — CG on (10.23) with products via J then J^T ; Chapter 7 transfers wholesale, simplified by $J^T J \succeq 0$ (no negative-curvature exit needed).

Convergence of Gauss–Newton (Theorem 10.1, and the honest local rate)

Theorem 10.1 (global)

If each r_j is Lipschitz continuously differentiable near the bounded level set, and the singular values of $J(x)$ are uniformly bounded away from zero ($\|J(x)z\| \geq \gamma\|z\|$, 10.28), then GN with Wolfe steps gives $J_k^T r_k = \nabla f_k \rightarrow 0$.

Proof shape: (10.25) and (10.28) give $\cos \theta_k \geq \gamma^2 / \bar{\beta}^2 > 0$ uniformly — then Zoutendijk. Rank-deficiency is the real threat: (10.23) stays solvable, but $\cos \theta_k$ loses its floor.

Local rate (10.30): read the constant

Near x^* , with $H(x) = \sum r_j \nabla^2 r_j$ the neglected term,

$$\|x_k + p_k^{\text{GN}} - x^*\| \lesssim \|[J^T J(x^*)]^{-1} H(x^*)\| \|x_k - x^*\| + O(\|x_k - x^*\|^2).$$

- ▶ $H(x^*) = 0$ (zero residual): **quadratic**.
- ▶ $\|[J^T J]^{-1} H\| \ll 1$ (small residual): fast linear, Newton-like in practice.
- ▶ $\|[J^T J]^{-1} H\| \approx 1$ or more (large residual / strong curvature): slow or **no local convergence**. The rate degrades *continuously* with residual size — there is no cliff, only a dial.

Levenberg–Marquardt: the same approximation, in a trust region

Same model, trust region instead of line search:

$$\min_p \frac{1}{2} \|J_k p + r_k\|^2 \quad \text{s.t. } \|p\| \leq \Delta_k \quad (10.31)$$

— historically the *progenitor* of Chapter 4, here presented as its special case. Fixes GN's weak spot: (near-)rank-deficient J_k .

Lemma 10.2 (= Theorem 4.1, with one condition for free)

p^{LM} solves (10.31) iff feasible and for some $\lambda \geq 0$:

$$(J^T J + \lambda I) p^{\text{LM}} = -J^T r, \quad \lambda(\Delta - \|p^{\text{LM}}\|) = 0. \quad (10.34)$$

Condition (4.8c), $B + \lambda I \succeq 0$, holds automatically — $J^T J \succeq 0$ already.

Implementation: still never form $J^T J$

(10.34a) is the normal equation of the augmented linear LS problem $\min_p \frac{1}{2} \left\| \begin{bmatrix} J \\ \sqrt{\lambda} I \end{bmatrix} p + \begin{bmatrix} r \\ 0 \end{bmatrix} \right\|^2$ (10.35) — QR it. Better: factor $J = QR$ once (10.37); each trial λ then costs only $\sim n^2/2$. Givens rotations to eliminate $\sqrt{\lambda} I$ (10.36–10.38). The Ch. 4 root-finder tunes λ to Δ ; safeguarding is trivial — any $\lambda > 0$ makes the system positive definite.

Convergence of LM, and the refrain

Theorem 10.3 (via Theorem 4.6)

$\eta \in (0, \frac{1}{4})$ in Algorithm 4.1, level set bounded, r_j Lipschitz continuously differentiable nearby, and each approximate subproblem solution achieving the Cauchy-fraction decrease (10.42) with $\|p_k\| \leq \gamma \Delta_k$. Then $\nabla f_k = J_k^T r_k \rightarrow 0$.

- ▶ No exact subproblem solves needed; **CG–Steihaug satisfies (10.42) automatically with $c_1 = \frac{1}{2}$** (its first iterate is the Cauchy point — Ch. 7).
- ▶ Locally: near a small-residual solution the bound goes inactive, steps become pure GN, and the rate is (10.30) again. LM buys robustness far away, not speed near the end.
- ▶ Large-residual problems degrade LM exactly as they degrade GN — same Hessian model.

The refrain, third verse

$(J^T J + \lambda I)p = -J^T r$: the LM step *is* the trust-region step *is* the Tikhonov-regularized linear LS solution. One object — Ch. 3's τI modification, Ch. 4's λ from the radius, Ch. 19's inertia correction — wearing a fourth name.

When residuals are large: hybrids and structured quasi-Newton

If $H(x^*) = \sum r_j \nabla^2 r_j$ is not negligible, GN/LM crawl. (Large residuals often mean the *model* is wrong — but you still must solve the fit to find that out.) Newton needs the $\nabla^2 r_j$; plain BFGS ignores the free $J^T J$ part.

Two repairs:

Hybrid (Fletcher–Xu)

Try the GN step; if it cuts f by a fixed factor, take it and reset $B_k \leftarrow J_k^T J_k$; otherwise take a quasi-Newton step with B_k and update by BFGS. Zero-residual problems $\rightarrow$ eventually pure GN (quadratic); large-residual $\rightarrow$ eventually pure BFGS (superlinear). Both limits right.

Structured quasi-Newton (Dennis–Gay–Welsch, NL2SOL)

Keep the exact free part; approximate *only the missing part*: $B_k = J_k^T J_k + S_k$, with S_k updated by a secant condition derived from per-residual secants:

$$S_{k+1} s_k = J_{k+1}^T r_{k+1} - J_k^T r_{k+1}$$

— the quasi-Newton principle of Ch. 6, applied surgically to the unknown term only.

Chapter 6's advice echoes: approximate what you cannot compute; never approximate what you are holding in your hand.

Orthogonal distance regression (§10.4, one slide)

Fixed-regressor assumed the t_j exact. If the abscissae are noisy too (**errors-in-variables**), vertical distances are the wrong penalty — use the (weighted) *shortest* distances to the curve:

$$\min_{x, \delta} \frac{1}{2} \sum_{j=1}^m \left[w_j^2 (\phi(x; t_j + \delta_j) - y_j)^2 + d_j^2 \delta_j^2 \right],$$

jointly over parameters $x \in \mathbb{R}^n$ and per-point shifts $\delta \in \mathbb{R}^m$.

- ▶ Formally a least-squares problem with $m + n$ unknowns and $2m$ residuals — looks m times bigger, but the δ_j couple only through their own residual pair: the Jacobian has an **arrow/block-diagonal structure**, and LM implementations exploit it so the cost stays comparable to ordinary LS.
- ▶ Implemented in ODRPACK; the statistical name is *total least squares* (linear case).
- ▶ Same lesson as (10.27) and the compact forms of Ch. 7: *look at the sparsity before declaring a problem big*.

Chapter 10 in one table

Method	Hessian model	Globalization	Local rate	Best regime
Gauss–Newton	$J^T J$	Wolfe line search	(10.30); quadratic if $r^*=0$	small residual, full rank
Levenberg–Marquardt	$J^T J$	trust region	same as GN	small residual, any rank
Hybrid (Fletcher–Xu)	$J^T J$ or BFGS	either	right limit both ways	residual size unknown
DGW / NL2SOL	$J^T J + S_k$	either	superlinear	large residual
Newton	full (10.5)	either	quadratic	$\nabla^2 r_j$ affordable

Linear solver	Note
Normal eqs. + Cholesky	fastest; κ^2 ; fine for well-conditioned $m \gg n$
QR	the default; κ , not κ^2
SVD	most robust; sensitivity via σ_i ; truncation = regularization
LSQR / CG	large sparse; products with J, J^T only

Two ideas to carry forward

$J^T J$ is the free half of the Hessian, and its quality is measured by the residuals. $(J^T J + \lambda I)p = -J^T r$: Levenberg–Marquardt, trust region, and Tikhonov are one object.

Homework

Derivations

- 1 Derive (10.4)–(10.5) from $f = \frac{1}{2}\|r\|^2$, and verify $\nabla^2 f = J^T J$ exactly in the linear case.
- 2 Prove the descent property (10.25), including the degenerate case $J_k p_k^{\text{GN}} = 0$.
- 3 Derive Lemma 10.2 from Theorem 4.1, checking why (4.8c) is automatic.
- 4 Show that (10.34a) is the normal equation of the augmented problem (10.35).
- 5 Derive the DGW secant equation $S_{k+1} s_k = J_{k+1}^T r_{k+1} - J_k^T r_{k+1}$ from per-residual secant conditions.

Implementation

6. Generate synthetic data from the model of Example 10.1 with Gaussian noise. Implement GN (with your Ch. 3 line search, step via QR of J_k) and LM (Algorithm 4.1 + (10.35)). Compare iterations and robustness from a distant start.
7. *The κ^2 experiment.* Build an ill-conditioned linear LS problem ($\kappa(J) \approx 10^7$); solve by normal equations + Cholesky and by QR; compare errors against an SVD reference solution.
8. *The residual dial.* Refit item 6 while scaling the noise up by factors of 10. Log the per-iteration contraction of $\|x_k - x^*\|$ for GN and for BFGS; find the noise level where BFGS starts winning — then check it against $\|[J^T J]^{-1} H\|$ at the solution.

Next: Ch. 11 — *nonlinear equations: the same structure with no objective to fall back on.*